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Portfolio Optimization Report: Comparative Analysis of Traditional MVO and Factor-Based Strategies

Introduction

This report summarizes a portfolio optimization study using historical data from 20 major US stocks spanning January 2010 to December 2024. The analysis compares two approaches:

- Traditional Mean-Variance Optimization (MVO) Strategy
- Factor-Based Investment Strategy

Both strategies were evaluated using key performance metrics to assess their effectiveness in balancing risk and return over the observed period [1].

Data and Methodology

Data Collection

- Historical monthly closing prices for 20 large-cap US stocks (e.g., AAPL, MSFT, GOOGL, AMZN, META, JPM, etc.) were sourced from Yahoo Finance.
- Monthly returns were computed for each stock, resulting in a dataset of 152 monthly periods [1].

Portfolio Strategies

- **Traditional MVO Strategy:** Utilizes Modern Portfolio Theory to optimize asset weights for maximum expected return at a given risk level, based on historical return and covariance data [2] [3].
- **Factor-Based Strategy:** Allocates assets based on exposure to specific risk factors (e.g., value, momentum, size), aiming to systematically capture risk premia identified in academic finance research [4] [5].

Performance Metrics

The following metrics were used to evaluate both strategies:

Metric	Traditional MVO	Factor-Based Strategy
Cumulative Return	14%	8%
CAGR	1%	0%
Sharpe Ratio	5.72	4.84

Metric	Traditional MVO	Factor-Based Strategy
Sortino Ratio	10.12	8.27
Max Drawdown	-2%	-1%
Longest Drawdown (days)	456	336
Gain/Pain Ratio	1.42	1.18
Profit Factor	2.42	2.18
Recovery Factor	6.97	10.6
Serenity Index	10.17	24.03

Analysis and Discussion

Traditional MVO Strategy

- **Risk-Return Profile:** Delivered higher cumulative return (14%) and a positive CAGR (1%), outperforming the factor-based approach on most risk-adjusted metrics.
- **Sharpe and Sortino Ratios:** High Sharpe (5.72) and Sortino (10.12) ratios indicate strong risk-adjusted performance.
- **Drawdowns:** Slightly higher maximum drawdown (-2%) and longer recovery period (456 days), suggesting more prolonged periods of underperformance during market stress.

Factor-Based Strategy

- **Risk-Return Profile:** Achieved lower cumulative return (8%) and flat CAGR, but with a lower maximum drawdown (-1%) and shorter recovery period (336 days).
- **Stability:** Higher Recovery Factor (10.6) and Serenity Index (24.03) suggest better resilience and smoother recovery after losses.
- **Risk Management:** The lower drawdown and higher resilience are consistent with the theoretical benefits of factor investing, which seeks to diversify risk exposures beyond traditional market-cap weighting[4] [5].

Comparative Insights

- **Return vs. Risk:** The MVO strategy offered higher returns but with slightly more risk and longer drawdown periods. The factor-based approach, while lagging in returns, provided a smoother ride with less severe and shorter drawdowns.
- **Suitability:** Investors prioritizing maximum returns and willing to accept longer recovery periods may prefer MVO. Those seeking stability and faster recovery from losses may find factor-based strategies more appealing.

Conclusion

Both portfolio optimization strategies demonstrated strengths and trade-offs:

- **Traditional MVO** is superior in terms of total and risk-adjusted returns over the sample period, but at the cost of deeper and longer drawdowns.
- **Factor-Based Strategy** excels in drawdown management and recovery, providing a more defensive profile at the expense of lower returns.

The choice between these approaches should align with the investor's risk tolerance, return objectives, and investment horizon. Combining elements from both strategies could potentially yield a more balanced risk-return profile [4] [5] [3].

Recommendations

- **For Return-Seeking Investors:** Prioritize traditional MVO, but monitor drawdown risk.
- **For Risk-Averse Investors:** Consider factor-based approaches for smoother performance and quicker recovery.
- **Hybrid Approach:** Explore multi-factor portfolios or blending MVO and factor-based allocations to optimize both return and risk management.

This analysis is based on historical data and past performance does not guarantee future results. Portfolio construction should always consider individual investor objectives and constraints.

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1. Falgun_240387_Finoptix-3.ipynb
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